Introduction

CISC 7404 - Decision Making

Steven Morad

University of Macau

Prerequisites 2
Grading 7
Resources 17
Cheating 19
Lecture Topics 27
What is Decision Making? 32
History of Decision Making 39
Decision Making and Deep Learning 52
Questions? 62
Homework 63

Prerequisites

Prerequisites

- Python numerical programming

Prerequisites

- Python numerical programming
- Deep learning

Prerequisites

- Python numerical programming
- Deep learning
- Statistics and probability

Prerequisites - Numerical Programming

You should know:

Prerequisites - Numerical Programming

You should know:

- Python loops, lists, dicts, etc

Prerequisites - Numerical Programming

You should know:

- Python loops, lists, dicts, etc
- Python objects, inheritance, functional programming, etc

Prerequisites - Numerical Programming

You should know:

- Python loops, lists, dicts, etc
- Python objects, inheritance, functional programming, etc
- `numpy` and `torch`

Prerequisites - Numerical Programming

You should know:

- Python loops, lists, dicts, etc
- Python objects, inheritance, functional programming, etc
- `numpy` and `torch`
  - Batch matrix multiply, elementwise product, sum, max, etc

Prerequisites - Numerical Programming

You should know:

- Python loops, lists, dicts, etc
- Python objects, inheritance, functional programming, etc
- `numpy` and `torch`
  - Batch matrix multiply, elementwise product, sum, max, etc
  - Multidimensonal tensors (i.e., shape)

Prerequisites - Numerical Programming

You should know:

- Python loops, lists, dicts, etc
- Python objects, inheritance, functional programming, etc
- `numpy` and `torch`
  - Batch matrix multiply, elementwise product, sum, max, etc
  - Multidimensonal tensors (i.e., shape)

Assignments in `jax/equinox`, similar to `torch`

Prerequisites - Numerical Programming

You should know:

- Python loops, lists, dicts, etc
- Python objects, inheritance, functional programming, etc
- `numpy` and `torch`
  - Batch matrix multiply, elementwise product, sum, max, etc
  - Multidimensonal tensors (i.e., shape)

Assignments in `jax/equinox`, similar to `torch`

If you do not know numerical programming, **you must learn immediately**: https://numpy.org/doc/stable/user/quickstart.html

E6-1102C

Prerequisites - Deep Learning

You should know:

Prerequisites - Deep Learning

You should know:

- How to construct a neural network in `torch`

Prerequisites - Deep Learning

You should know:

- How to construct a neural network in `torch`
- Classification and regression losses

Prerequisites - Deep Learning

You should know:

- How to construct a neural network in `torch`
- Classification and regression losses
- Optimization/SGD

Prerequisites - Deep Learning

You should know:

- How to construct a neural network in `torch`
- Classification and regression losses
- Optimization/SGD
- How to train a neural network

Prerequisites - Deep Learning

You should know:

- How to construct a neural network in `torch`
- Classification and regression losses
- Optimization/SGD
- How to train a neural network
- Multilayer perceptrons

Prerequisites - Deep Learning

You should know:

- How to construct a neural network in `torch`
- Classification and regression losses
- Optimization/SGD
- How to train a neural network
- Multilayer perceptrons
- Convolutional networks

Prerequisites - Deep Learning

You should know:

- How to construct a neural network in `torch`
- Classification and regression losses
- Optimization/SGD
- How to train a neural network
- Multilayer perceptrons
- Convolutional networks

If you do not, review the deep learning slides: https://github.com/smorad/um_cisc_7026

Prerequisites - Statistics

You should know:

Prerequisites - Statistics

You should know:

- Random variables

Prerequisites - Statistics

You should know:

- Random variables
- Probability density and mass functions (PDF, PMF)

Prerequisites - Statistics

You should know:

- Random variables
- Probability density and mass functions (PDF, PMF)
- Conditional and unconditional probabilities

Prerequisites - Statistics

You should know:

- Random variables
- Probability density and mass functions (PDF, PMF)
- Conditional and unconditional probabilities
- Expectations/expected values

Prerequisites - Statistics

You should know:

- Random variables
- Probability density and mass functions (PDF, PMF)
- Conditional and unconditional probabilities
- Expectations/expected values

Question: What does $P(X = x)$ mean?

Prerequisites - Statistics

You should know:

- Random variables
- Probability density and mass functions (PDF, PMF)
- Conditional and unconditional probabilities
- Expectations/expected values

Question: What does $P(X = x)$ mean?

Answer: Probability of random variable X taking on a value of x

Prerequisites - Statistics

You should know:

- Random variables
- Probability density and mass functions (PDF, PMF)
- Conditional and unconditional probabilities
- Expectations/expected values

Question: What does $P(X = x)$ mean?

Answer: Probability of random variable X taking on a value of x

If you did not know this, you should review!

Grading

Grading

All of the following information exists on Moodle

Grading

All of the following information exists on Moodle

- Syllabus

Grading

All of the following information exists on Moodle

- Syllabus
- Final project guidelines

Grading

- Exams 30%

Grading

- Exams 30%
- Assignments 30%

Grading

- Exams 30%
- Assignments 30%
- Final Project 30%

Grading

- Exams 30%
- Assignments 30%
- Final Project 30%
- Participation 10%

Grading - Exams

- I will tell you at least 1 week before exam

Grading - Exams

- I will tell you at least 1 week before exam
- Expect 3 exams

Grading - Exams

- I will tell you at least 1 week before exam
- Expect 3 exams
- I will drop your lowest exam score

Grading - Exams

- I will tell you at least 1 week before exam
- Expect 3 exams
- I will drop your lowest exam score

Example 1: Exam 1: 70%, Exam 2: 80%, Exam 3: 60%

Grading - Exams

- I will tell you at least 1 week before exam
- Expect 3 exams
- I will drop your lowest exam score

Example 1: Exam 1: 70%, Exam 2: 80%, Exam 3: 60%

Final exam score: (70 + 80) / 2 = 75%

Grading - Exams

- I will tell you at least 1 week before exam
- Expect 3 exams
- I will drop your lowest exam score

Example 1: Exam 1: 70%, Exam 2: 80%, Exam 3: 60%

Final exam score: (70 + 80) / 2 = 75%

Example 2: Exam 1: 90%, Exam 2: (sick) 0%, Exam 3: 70%

Grading - Exams

- I will tell you at least 1 week before exam
- Expect 3 exams
- I will drop your lowest exam score

Example 1: Exam 1: 70%, Exam 2: 80%, Exam 3: 60%

Final exam score: (70 + 80) / 2 = 75%

Example 2: Exam 1: 90%, Exam 2: (sick) 0%, Exam 3: 70%

Final exam score: (90 + 70) / 2 = 80%

Grading - Exams

- I will tell you at least 1 week before exam
- Expect 3 exams
- I will drop your lowest exam score

Example 1: Exam 1: 70%, Exam 2: 80%, Exam 3: 60%

Final exam score: (70 + 80) / 2 = 75%

Example 2: Exam 1: 90%, Exam 2: (sick) 0%, Exam 3: 70%

Final exam score: (90 + 70) / 2 = 80%

Question: What if you are sick for two examzes?

Grading - Exams

- I will tell you at least 1 week before exam
- Expect 3 exams
- I will drop your lowest exam score

Example 1: Exam 1: 70%, Exam 2: 80%, Exam 3: 60%

Final exam score: (70 + 80) / 2 = 75%

Example 2: Exam 1: 90%, Exam 2: (sick) 0%, Exam 3: 70%

Final exam score: (90 + 70) / 2 = 80%

Question: What if you are sick for two examzes? Only one exam dropped, other exam is zero

Grading - Assignments

- Programming

Grading - Assignments

- Programming
- Expect 3 assignments

Grading - Assignments

- Programming
- Expect 3 assignments
- We will use Google Colab: https://colab.research.google.com

Late assignment policy:

Grading - Assignments

- Programming
- Expect 3 assignments
- We will use Google Colab: https://colab.research.google.com

Late assignment policy:

- 0-1 day late (-15%)

Grading - Assignments

- Programming
- Expect 3 assignments
- We will use Google Colab: https://colab.research.google.com

Late assignment policy:

- 0-1 day late (-15%)
- 1-2 days late (-30%)

Grading - Assignments

- Programming
- Expect 3 assignments
- We will use Google Colab: https://colab.research.google.com

Late assignment policy:

- 0-1 day late (-15%)
- 1-2 days late (-30%)
- 2-3 days late (-50%)

Grading - Assignments

- Programming
- Expect 3 assignments
- We will use Google Colab: https://colab.research.google.com

Late assignment policy:

- 0-1 day late (-15%)
- 1-2 days late (-30%)
- 2-3 days late (-50%)
- 3+ days late (-100%)

Grading - Final Project

- Form groups of 3-5

Grading - Final Project

Grading - Final Project

- Form groups of 3-5
- Can do any project related to decision making

Grading - Final Project

- Form groups of 3-5
- Can do any project related to decision making

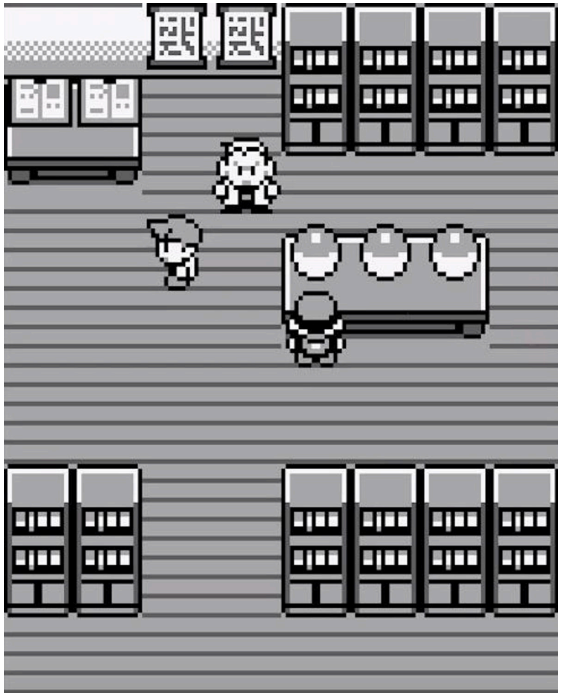
Starcraft

Grading - Final Project

- Form groups of 3-5
- Can do any project related to decision making

Starcraft

Pokemon

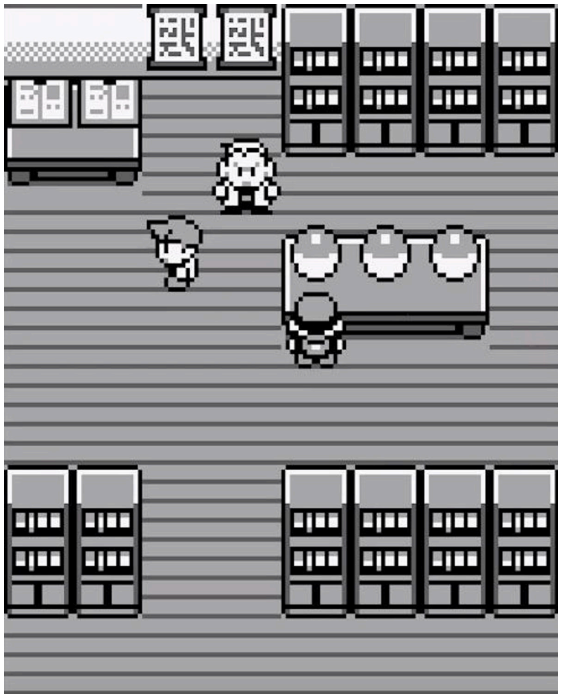


Grading - Final Project

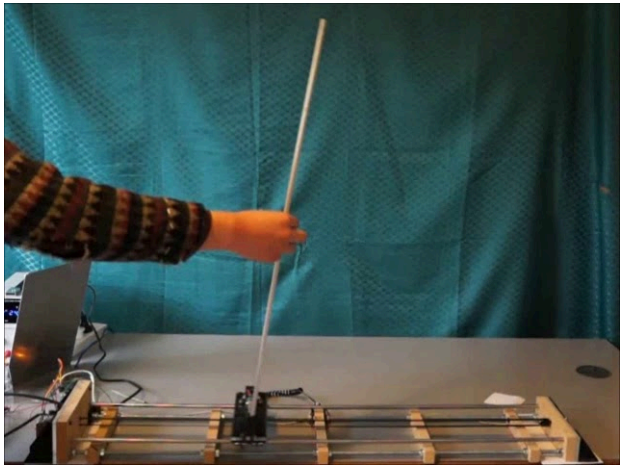
- Form groups of 3-5
- Can do any project related to decision making

Starcraft

Pokemon



Robot Control



Grading - Final Project

See Moodle for final project specification

Grading - Final Project

See Moodle for final project specification

- 30% implementation

Grading - Final Project

See Moodle for final project specification

- 30% implementation
- 30% writeup and video

Grading - Final Project

See Moodle for final project specification

- 30% implementation
- 30% writeup and video
- 30% difficulty modifier

Grading - Final Project

See Moodle for final project specification

- 30% implementation
- 30% writeup and video
- 30% difficulty modifier
  - Trivial projects will get 0 difficult points!

Grading - Participation

Significance

Despite active learning being recognized as a superior method of instruction in the classroom, a major recent survey found that most college STEM instructors still choose traditional teaching methods. This article addresses the long-standing question of why students and faculty remain resistant to active learning. Comparing passive lectures with active learning using a randomized experimental approach and identical course materials, we find that students in the active classroom learn more, but they feel like they learn less. We show that this negative correlation is caused in part by the increased cognitive effort required during active learning. Faculty who adopt active learning are encouraged to intervene and address this misperception, and we describe a successful example of such an intervention.

Grading - Participation

Participation: I want this class to be interactive

Grading - Participation

Participation: I want this class to be interactive

Participation is **asking** or **answering** questions during lecture

Grading - Participation

Participation: I want this class to be interactive

Participation is **asking** or **answering** questions during lecture

To encourage you, your particiipation grade depends on interacting

Grading - Participation

Participation: I want this class to be interactive

Participation is **asking** or **answering** questions during lecture

To encourage you, your particiipation grade depends on interacting

- Class participation (50%)

Grading - Participation

Participation: I want this class to be interactive

Participation is **asking** or **answering** questions during lecture

To encourage you, your particiipation grade depends on interacting

- Class participation (50%)
- Individual participation (50%)

Grading - Participation

Participation: I want this class to be interactive

Participation is **asking** or **answering** questions during lecture

To encourage you, your particiipation grade depends on interacting

- Class participation (50%)
- Individual participation (50%)

You MUST ask or answer questions during lecture

Grading - Participation

Participation: I want this class to be interactive

Participation is **asking** or **answering** questions during lecture

To encourage you, your particiipation grade depends on interacting

- Class participation (50%)
- Individual participation (50%)

You MUST ask or answer questions during lecture

- If not, you will receive **0 for individual participation**

Resources

Resources

Office Hours: Moday 14:00 AM - 16:00 PM

Resources

Office Hours: Moday 14:00 AM - 16:00 PM

Textbook: http://incompleteideas.net/book/the-book-2nd.html

Resources

Office Hours: Moday 14:00 AM - 16:00 PM

Textbook: http://incompleteideas.net/book/the-book-2nd.html

- Syllabus lists textbook chapter for each lecture

Resources

Office Hours: Moday 14:00 AM - 16:00 PM

Textbook: http://incompleteideas.net/book/the-book-2nd.html

- Syllabus lists textbook chapter for each lecture

Last year lectures

Resources

Office Hours: Moday 14:00 AM - 16:00 PM

Textbook: http://incompleteideas.net/book/the-book-2nd.html

- Syllabus lists textbook chapter for each lecture

Last year lectures

- https://github.com/smorad/um_cisc_7404

Cheating

Cheating

Question: What is cheating?

Cheating

Question: What is cheating?

Answer:

Cheating

Question: What is cheating?

Answer:

- Copying assignment or exam from another student

Cheating

Question: What is cheating?

Answer:

- Copying assignment or exam from another student
- Having notes, laptop, or phone during exam/exam

Cheating

Question: What is cheating?

Answer:

- Copying assignment or exam from another student
- Having notes, laptop, or phone during exam/exam
- Turning in LLM-output

Cheating

I am serious about cheating

Cheating

I am serious about cheating

- Ask students who were in my Introduction to Deep Learning course

Cheating

I am serious about cheating

- Ask students who were in my Introduction to Deep Learning course
- Caught multiple students last term

Cheating

I am serious about cheating

- Ask students who were in my Introduction to Deep Learning course
- Caught multiple students last term
  - One student lost job offer

Cheating

I am serious about cheating

- Ask students who were in my Introduction to Deep Learning course
- Caught multiple students last term
  - One student lost job offer

Any student caught cheating:

Cheating

I am serious about cheating

- Ask students who were in my Introduction to Deep Learning course
- Caught multiple students last term
  - One student lost job offer

Any student caught cheating:

- Receives 0 (fail) in course

Cheating

I am serious about cheating

- Ask students who were in my Introduction to Deep Learning course
- Caught multiple students last term
  - One student lost job offer

Any student caught cheating:

- Receives 0 (fail) in course
- Reported to head of department

Cheating

I am serious about cheating

- Ask students who were in my Introduction to Deep Learning course
- Caught multiple students last term
  - One student lost job offer

Any student caught cheating:

- Receives 0 (fail) in course
- Reported to head of department

It is not worth cheating, do your best and you will get partial credit

Cheating

Secret: After you graduate nobody will care about your grade

Cheating

Secret: After you graduate nobody will care about your grade

For AI jobs, you will do 5 hours of in-person interviews

Cheating

Secret: After you graduate nobody will care about your grade

For AI jobs, you will do 5 hours of in-person interviews

You will write on a whiteboard in front of interviewers

Cheating

Secret: After you graduate nobody will care about your grade

For AI jobs, you will do 5 hours of in-person interviews

You will write on a whiteboard in front of interviewers

There is nobody to copy and no LLM to help

Cheating

Secret: After you graduate nobody will care about your grade

For AI jobs, you will do 5 hours of in-person interviews

You will write on a whiteboard in front of interviewers

There is nobody to copy and no LLM to help

To get the job, you must **understand** the material

Cheating

Secret: After you graduate nobody will care about your grade

For AI jobs, you will do 5 hours of in-person interviews

You will write on a whiteboard in front of interviewers

There is nobody to copy and no LLM to help

To get the job, you must **understand** the material

Your degree will be useless without the knowledge

Cheating

Secret: After you graduate nobody will care about your grade

For AI jobs, you will do 5 hours of in-person interviews

You will write on a whiteboard in front of interviewers

There is nobody to copy and no LLM to help

To get the job, you must **understand** the material

Your degree will be useless without the knowledge

I want you to **learn the material** so you succeed in life

Cheating

Machine Learning Engineer Interview

16 May 2024 •••

Anonymous interview candidate

✕ No offer — Neutral experience ✕ Difficult interview

Application

I interviewed at Baidu in 16/5/2024

Interview

it has 5 rounds of interviews, each of them is very long. I only managed to get to the second round with technical interview, and I failed. It's not a pleasent experience

Interview questions [1]

Question 1

describle tree algorithm and wirte in python

Answer question →

Helpful Share

Cheating

Machine Learning Engineer Interview

Aug 27, 2023 •••

Anonymous Interview Candidate

Beijing, Beijing

— Declined offer ✓ Positive experience ✗ Difficult interview

Application

I applied online. I interviewed at ByteDance (Beijing, Beijing) in 8/27/2023

Interview

machine learning knowledge, code. leetcode (medium, hard with python), AUC definition, method on dealing with long tailed data and causal inference, machine learning question and deep learning method such as attention mechaism.

Interview questions [1]

Question 1

machine learning knowledge, code. leetcode, AUC definition, method on dealin...

read more

Answer question →

Cheating - LLMs

Question: Can you use LLMs in class?

Cheating - LLMs

Question: Can you use LLMs in class?

You can ask LLMs for help, but **do not turn in LLM output**

Cheating - LLMs

Question: Can you use LLMs in class?

You can ask LLMs for help, but **do not turn in LLM output**

Ok: LLM, is ____ a correct translation of ____

Cheating - LLMs

Question: Can you use LLMs in class?

You can ask LLMs for help, but **do not turn in LLM output**

Ok: LLM, is ____ a correct translation of ____

Ok: LLM, is ____ correct grammar?

Cheating - LLMs

Question: Can you use LLMs in class?

You can ask LLMs for help, but **do not turn in LLM output**

Ok: LLM, is ____ a correct translation of ____

Ok: LLM, is ____ correct grammar?

Cheating: LLM, answer the following homework question ____

Cheating - LLMs

Question: Can you use LLMs in class?

You can ask LLMs for help, but **do not turn in LLM output**

Ok: LLM, is ____ a correct translation of ____

Ok: LLM, is ____ correct grammar?

Cheating: LLM, answer the following homework question ____

Ok: LLM, why does my code raise `AttributeError`?

Cheating - LLMs

Question: Can you use LLMs in class?

You can ask LLMs for help, but **do not turn in LLM output**

Ok: LLM, is ____ a correct translation of ____

Ok: LLM, is ____ correct grammar?

Cheating: LLM, answer the following homework question ____

Ok: LLM, why does my code raise `AttributeError`?

Ok: LLM, why does my Q function return large values?

Cheating - LLMs

Question: Can you use LLMs in class?

You can ask LLMs for help, but **do not turn in LLM output**

Ok: LLM, is ____ a correct translation of ____

Ok: LLM, is ____ correct grammar?

Cheating: LLM, answer the following homework question ____

Ok: LLM, why does my code raise `AttributeError`?

Ok: LLM, why does my Q function return large values?

Cheating: LLM, implement the policy gradient algorithm in pytorch

Cheating - LLMs

It is hard to prove LLM usage in assignments

Cheating - LLMs

It is hard to prove LLM usage in assignments

- Detectors have some false positives

Cheating - LLMs

It is hard to prove LLM usage in assignments

- Detectors have some false positives
- Unlikely strong enough evidence to fail course

Cheating - LLMs

It is hard to prove LLM usage in assignments

- Detectors have some false positives
- Unlikely strong enough evidence to fail course
- If I think code is LLM generated, I will give zero for assignment

Lecture Topics

Lecture Topics

- Basics

Lecture Topics

- Basics
- Modern Methods

Lecture Topics

- Basics
- Modern Methods
- Active Research

Lecture Topics - Basics

- Bandits

Lecture Topics - Basics

- Bandits
- Decision Processes

Lecture Topics - Basics

- Bandits
- Decision Processes
- Trajectory Optimization

Lecture Topics - Basics

- Bandits
- Decision Processes
- Trajectory Optimization
- Value Iteration

Lecture Topics - Basics

- Bandits
- Decision Processes
- Trajectory Optimization
- Value Iteration
- Policy Gradient

Lecture Topics - Basics

- Bandits
- Decision Processes
- Trajectory Optimization
- Value Iteration
- Policy Gradient

Lecture Topics - Modern Methods

We will study modern methods (mostly actor-critic)

Lecture Topics - Modern Methods

We will study modern methods (mostly actor-critic)

- Deep Q Learning

Lecture Topics - Modern Methods

We will study modern methods (mostly actor-critic)

- Deep Q Learning
- Advantage Actor Critic

Lecture Topics - Modern Methods

We will study modern methods (mostly actor-critic)

- Deep Q Learning
- Advantage Actor Critic
- Trust Region Policy Optimization

Lecture Topics - Modern Methods

We will study modern methods (mostly actor-critic)

- Deep Q Learning
- Advantage Actor Critic
- Trust Region Policy Optimization
- Proximal Policy Optimization

Lecture Topics - Modern Methods

We will study modern methods (mostly actor-critic)

- Deep Q Learning
- Advantage Actor Critic
- Trust Region Policy Optimization
- Proximal Policy Optimization
- Deep Deterministic Policy Gradient

Lecture Topics - Modern Methods

We will study modern methods (mostly actor-critic)

- Deep Q Learning
- Advantage Actor Critic
- Trust Region Policy Optimization
- Proximal Policy Optimization
- Deep Deterministic Policy Gradient
- Soft Actor Critic

Lecture Topics - Active Research

We will study offline methods

Lecture Topics - Active Research

We will study offline methods

- Behavioral Cloning

Lecture Topics - Active Research

We will study offline methods

- Behavioral Cloning
- Monotonic Advantage Re-Weighted Imitation Learning

Lecture Topics - Active Research

We will study offline methods

- Behavioral Cloning
- Monotonic Advantage Re-Weighted Imitation Learning
- Batch Constrained Q Learning
- Conservative Q Learning

Lecture Topics - Active Research

We will study offline methods

- Behavioral Cloning
- Monotonic Advantage Re-Weighted Imitation Learning
- Batch Constrained Q Learning
- Conservative Q Learning

We will also study some LLM topics

- Learning from Preferences

Lecture Topics - Active Research

We will study offline methods

- Behavioral Cloning
- Monotonic Advantage Re-Weighted Imitation Learning
- Batch Constrained Q Learning
- Conservative Q Learning

We will also study some LLM topics

- Learning from Preferences
- Reinforcement Learning from Human Feedback

Lecture Topics - Active Research

We will study offline methods

- Behavioral Cloning
- Monotonic Advantage Re-Weighted Imitation Learning
- Batch Constrained Q Learning
- Conservative Q Learning

We will also study some LLM topics

- Learning from Preferences
- Reinforcement Learning from Human Feedback
- Group-Relative Policy Optimization

What is Decision Making?

What is Decision Making?

In this course, we will focus primarily on reinforcement learning

What is Decision Making?

In this course, we will focus primarily on reinforcement learning

But reinforcement learning is a method, not a problem

What is Decision Making?

In this course, we will focus primarily on reinforcement learning

But reinforcement learning is a method, not a problem

The problem is **decision making**

What is Decision Making?

In this course, we will focus primarily on reinforcement learning

But reinforcement learning is a method, not a problem

The problem is **decision making**

In this course, we will learn how to make good decisions

What is Decision Making?

Question: What is decision making?

What is Decision Making?

Question: What is decision making?

It depends, each field has their own definition

What is Decision Making?

Question: What is decision making?

It depends, each field has their own definition

- Philosophy

What is Decision Making?

Question: What is decision making?

It depends, each field has their own definition

- Philosophy
- Cognitive science

What is Decision Making?

Question: What is decision making?

It depends, each field has their own definition

- Philosophy
- Cognitive science
- Economics

What is Decision Making?

Question: What is decision making?

It depends, each field has their own definition

- Philosophy
- Cognitive science
- Economics
- Machine learning

What is Decision Making?

Question: What is decision making?

It depends, each field has their own definition

- Philosophy
- Cognitive science
- Economics
- Machine learning

Answer: Given information, make a choice that impacts the world

What is Decision Making?

Question: Why should we care about decision making?

What is Decision Making?

Question: Why should we care about decision making?

Everything in life is a decision

What is Decision Making?

Question: Why should we care about decision making?

Everything in life is a decision

- Do I eat dumplings or noodles?

What is Decision Making?

Question: Why should we care about decision making?

Everything in life is a decision

- Do I eat dumplings or noodles?
- What time should I leave for class?

What is Decision Making?

Question: Why should we care about decision making?

Everything in life is a decision

- Do I eat dumplings or noodles?
- What time should I leave for class?
- Should I go to school or find a job?

What is Decision Making?

Question: Why should we care about decision making?

Everything in life is a decision

- Do I eat dumplings or noodles?
- What time should I leave for class?
- Should I go to school or find a job?
- Should I dump my boyfriend?

What is Decision Making?

Question: Why should we care about decision making?

Everything in life is a decision

- Do I eat dumplings or noodles?
- What time should I leave for class?
- Should I go to school or find a job?
- Should I dump my boyfriend?
- Where should I live?

What is Decision Making?

Question: Why should we care about decision making?

Everything in life is a decision

- Do I eat dumplings or noodles?
- What time should I leave for class?
- Should I go to school or find a job?
- Should I dump my boyfriend?
- Where should I live?
- What should we use taxes for?

What is Decision Making?

Humans are decision making machines – it is all we do!

What is Decision Making?

Humans are decision making machines – it is all we do!

We can represent life as a series of decisions

What is Decision Making?

Humans are decision making machines – it is all we do!

We can represent life as a series of decisions

What we do defines who we are

What is Decision Making?

Humans are decision making machines – it is all we do!

We can represent life as a series of decisions

What we do defines who we are

"All we have to decide is what to do with the time that is given to us"

What is Decision Making?

Humans are decision making machines – it is all we do!

We can represent life as a series of decisions

What we do defines who we are

"All we have to decide is what to do with the time that is given to us"

To study decision making is to study ourselves

What is Decision Making?

Humans are decision making machines – it is all we do!

We can represent life as a series of decisions

What we do defines who we are

"All we have to decide is what to do with the time that is given to us"

To study decision making is to study ourselves

If we learn to make better decisions, we can lead better lives

What is Decision Making?

In this course, we focus on **optimal** decision making

What is Decision Making?

In this course, we focus on **optimal** decision making

Make the best possible decision with the information we have

What is Decision Making?

In this course, we focus on **optimal** decision making

Make the best possible decision with the information we have

We will find methods that **guarantee** optimal decisions

What is Decision Making?

In this course, we focus on **optimal** decision making

Make the best possible decision with the information we have

We will find methods that **guarantee** optimal decisions

With these methods, we can create optimal decision making machines

What is Decision Making?

In this course, we focus on **optimal** decision making

Make the best possible decision with the information we have

We will find methods that **guarantee** optimal decisions

With these methods, we can create optimal decision making machines

With an optimal decision making machine, you can create:

What is Decision Making?

In this course, we focus on **optimal** decision making

Make the best possible decision with the information we have

We will find methods that **guarantee** optimal decisions

With these methods, we can create optimal decision making machines

With an optimal decision making machine, you can create:

- Best possible doctor (which medicine to give?)

What is Decision Making?

In this course, we focus on **optimal** decision making

Make the best possible decision with the information we have

We will find methods that **guarantee** optimal decisions

With these methods, we can create optimal decision making machines

With an optimal decision making machine, you can create:

- Best possible doctor (which medicine to give?)
- Best possible lawyer (what to argue?)

What is Decision Making?

In this course, we focus on **optimal** decision making

Make the best possible decision with the information we have

We will find methods that **guarantee** optimal decisions

With these methods, we can create optimal decision making machines

With an optimal decision making machine, you can create:

- Best possible doctor (which medicine to give?)
- Best possible lawyer (what to argue?)
- Best possible scientist (what to research?)

What is Decision Making?

In this course, we focus on **optimal** decision making

Make the best possible decision with the information we have

We will find methods that **guarantee** optimal decisions

With these methods, we can create optimal decision making machines

With an optimal decision making machine, you can create:

- Best possible doctor (which medicine to give?)
- Best possible lawyer (what to argue?)
- Best possible scientist (what to research?)

If the machine understands **why** it makes decisions, is it conscious?

What is Decision Making?

Let us discuss the history of decision making to better understand it

History of Decision Making

History of Decision Making

Question: Who was the first to apply decision making algorithms?

History of Decision Making

Question: Who was the first to apply decision making algorithms?

3.5 GYA: Single cell organism

History of Decision Making

Question: Who was the first to apply decision making algorithms?

3.5 GYA: Single cell organism

Decides to move away from danger and move towards food

History of Decision Making

Question: Who was the first to apply decision making algorithms?

3.5 GYA: Single cell organism

Decides to move away from danger and move towards food

Decision making is necessary for life

History of Decision Making

History of Decision Making

200 kYA: Humanoid hunter-gatherers develop more complex decision making capabilities

History of Decision Making

200 kYA: Humanoid hunter-gatherers develop more complex decision making capabilities

Sequence of decisions to make fire

History of Decision Making

200 kYA: Humanoid hunter-gatherers develop more complex decision making capabilities

Sequence of decisions to make fire

Hunt animals or plant crops?

History of Decision Making

500 BCE: Humans begin to study decision making

History of Decision Making

500 BCE: Humans begin to study decision making

Sun Tzu studies and writes about zero-sum decision making

History of Decision Making

500 BCE: Humans begin to study decision making

Sun Tzu studies and writes about zero-sum decision making

E.g., "Attack where he is unprepared; appear where you are not expected."

History of Decision Making

400 BCE: Aristotle creates the earliest recorded framework for decision making

History of Decision Making

400 BCE: Aristotle creates the earliest recorded framework for decision making

Syllogistic logic and deductive reasoning from axioms

History of Decision Making

400 BCE: Aristotle creates the earliest recorded framework for decision making

Syllogistic logic and deductive reasoning from axioms

Axiom 1: All philosophers prioritize knowledge over leisure

History of Decision Making

400 BCE: Aristotle creates the earliest recorded framework for decision making

Syllogistic logic and deductive reasoning from axioms

Axiom 1: All philosophers prioritize knowledge over leisure

Axiom 2: I am a philosopher

History of Decision Making

400 BCE: Aristotle creates the earliest recorded framework for decision making

Syllogistic logic and deductive reasoning from axioms

Axiom 1: All philosophers prioritize knowledge over leisure

Axiom 2: I am a philosopher

Decision: I must attend lecture instead of the party

History of Decision Making

1654: Pascal formalizes decision making under uncertainty with "Pascal's Wager"

History of Decision Making

1654: Pascal formalizes decision making under uncertainty with "Pascal's Wager"

Premise: You are in bed, about to die. Should you believe in God?

History of Decision Making

1654: Pascal formalizes decision making under uncertainty with "Pascal's Wager"

Premise: You are in bed, about to die. Should you believe in God?

	Believe	Do not believe
God exists	Good	Bad
God does not exist	Neutral	Neutral

History of Decision Making

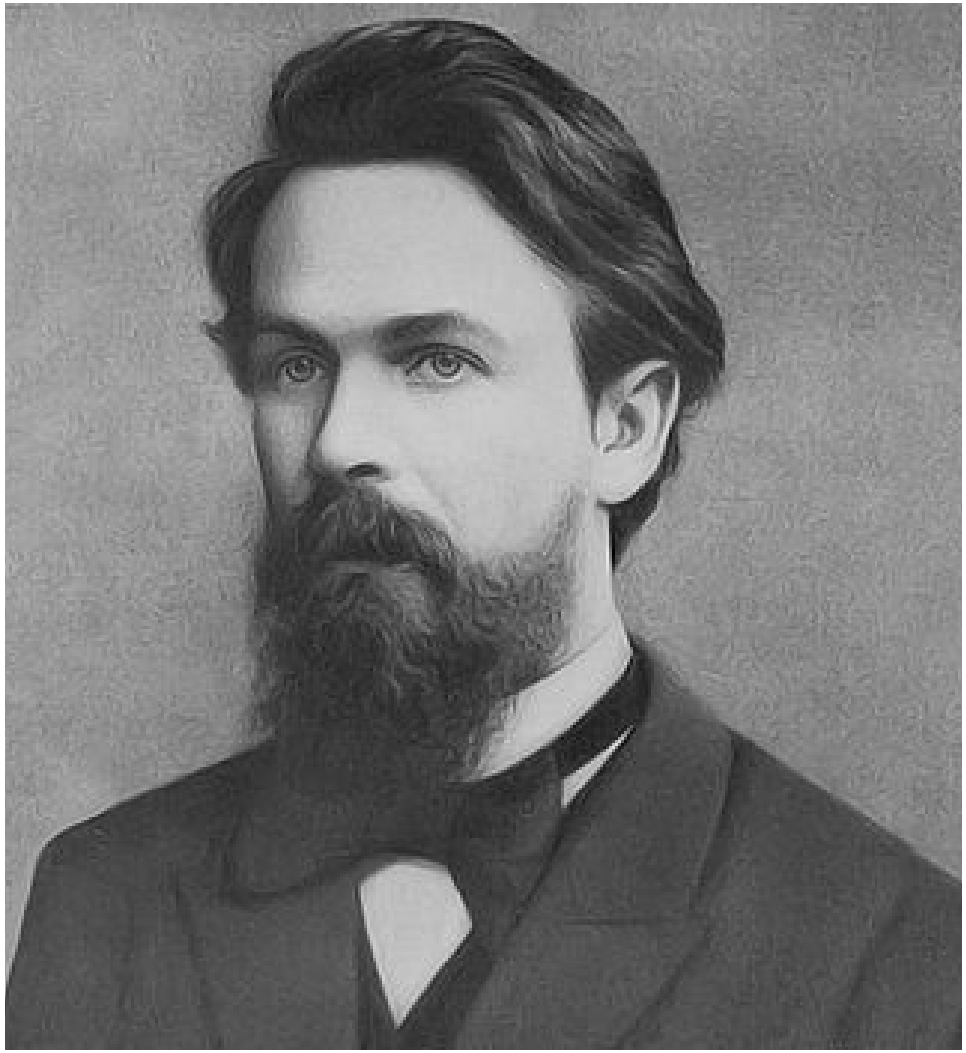
1654: Pascal formalizes decision making under uncertainty with "Pascal's Wager"

Premise: You are in bed, about to die. Should you believe in God?

	Believe	Do not believe
God exists	Good	Bad
God does not exist	Neutral	Neutral

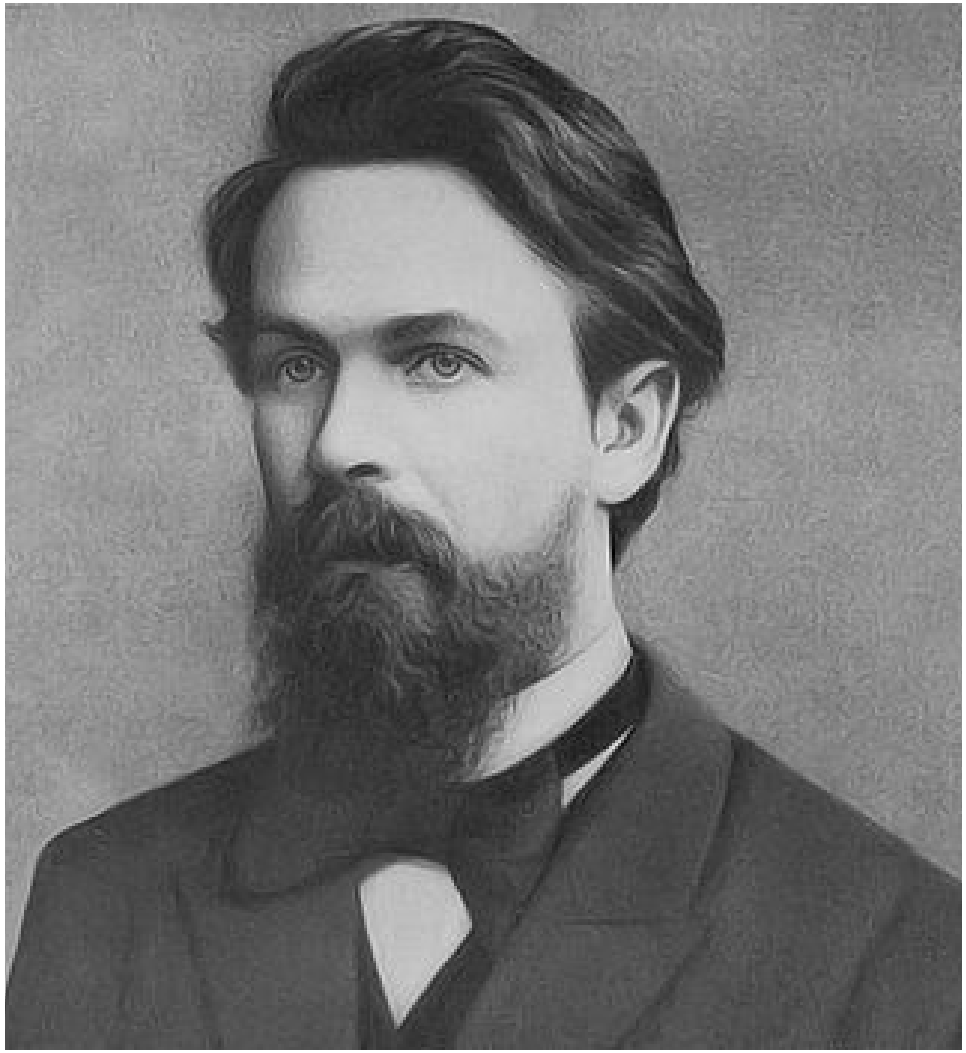
Note: I am not religious

History of Decision Making



1906: Markov discovers Markov processes

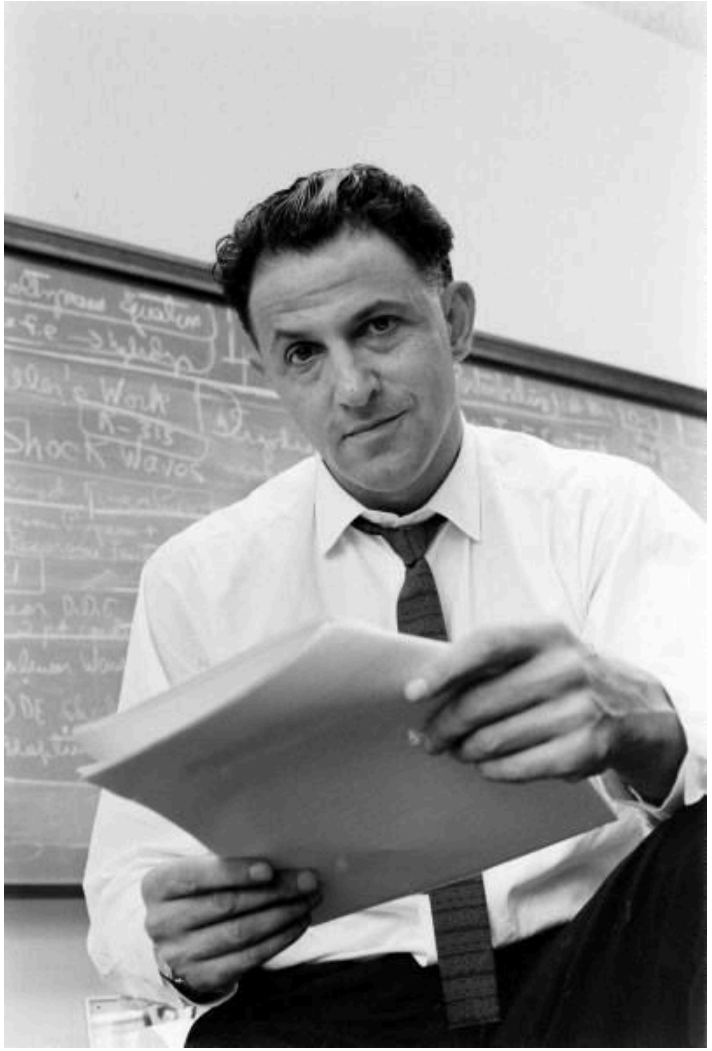
History of Decision Making



1906: Markov discovers Markov processes

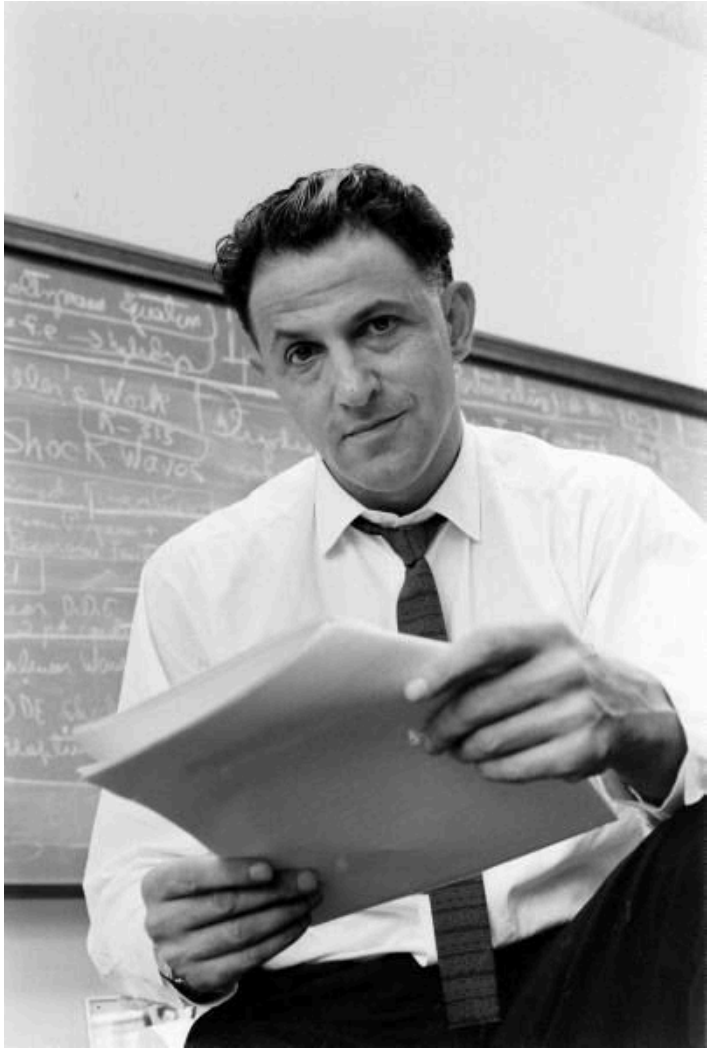
Modern decision making relies on Markov processes

History of Decision Making



1953: Bellman discovers dynamic programming

History of Decision Making



1953: Bellman discovers dynamic programming

Gives us the **Bellman equation**, the basis for optimal decision making

History of Decision Making

1983: Sutton solves the Bellman equation using neural networks

History of Decision Making

1983: Sutton solves the Bellman equation using neural networks

Combines reinforcement learning and neural networks

History of Decision Making

1983: Sutton solves the Bellman equation using neural networks

Combines reinforcement learning and neural networks

He is still alive and will answer your emails

History of Decision Making

1983: Sutton solves the Bellman equation using neural networks

Combines reinforcement learning and neural networks

He is still alive and will answer your emails

We use his textbook: *An Introduction to Reinforcement Learning*

History of Decision Making

1997: DeepBlue beats world champion Kasparov at chess

History of Decision Making

1997: DeepBlue beats world champion Kasparov at chess

People start to pay attention to decision making machines

History of Decision Making

1997: DeepBlue beats world champion Kasparov at chess

People start to pay attention to decision making machines

Chess AIs play each other because humans are too easy

History of Decision Making

1997: DeepBlue beats world champion Kasparov at chess

People start to pay attention to decision making machines

Chess AIs play each other because humans are too easy

https://www.youtube.com/watch?v=KF6sLCeBj0s

History of Decision Making

2016: AlphaGo beats world champion Sedol at Go

History of Decision Making

2016: AlphaGo beats world champion Sedol at Go

https://www.youtube.com/watch?v=tXlM99xPQC8

History of Decision Making

2018: OpenAI Five beats world champions at Dota2

History of Decision Making

2018: OpenAI Five beats world champions at Dota2

https://www.youtube.com/watch?v=eHipy_j29Xw

History of Decision Making

2020-2025: All frontier LLMs trained using reinforcement learning

History of Decision Making

2020-2025: All frontier LLMs trained using reinforcement learning

OpenAI, Baidu, Anthropic, DeepSeek, Google, Alibaba, ...

What will happen in 2026?

Decision Making and Deep Learning

Decision Making and Deep Learning

We will formally define decision making and reinforcement learning later in the course

Decision Making and Deep Learning

We will formally define decision making and reinforcement learning later in the course

For now, I want to clarify decision making in the context of machine learning

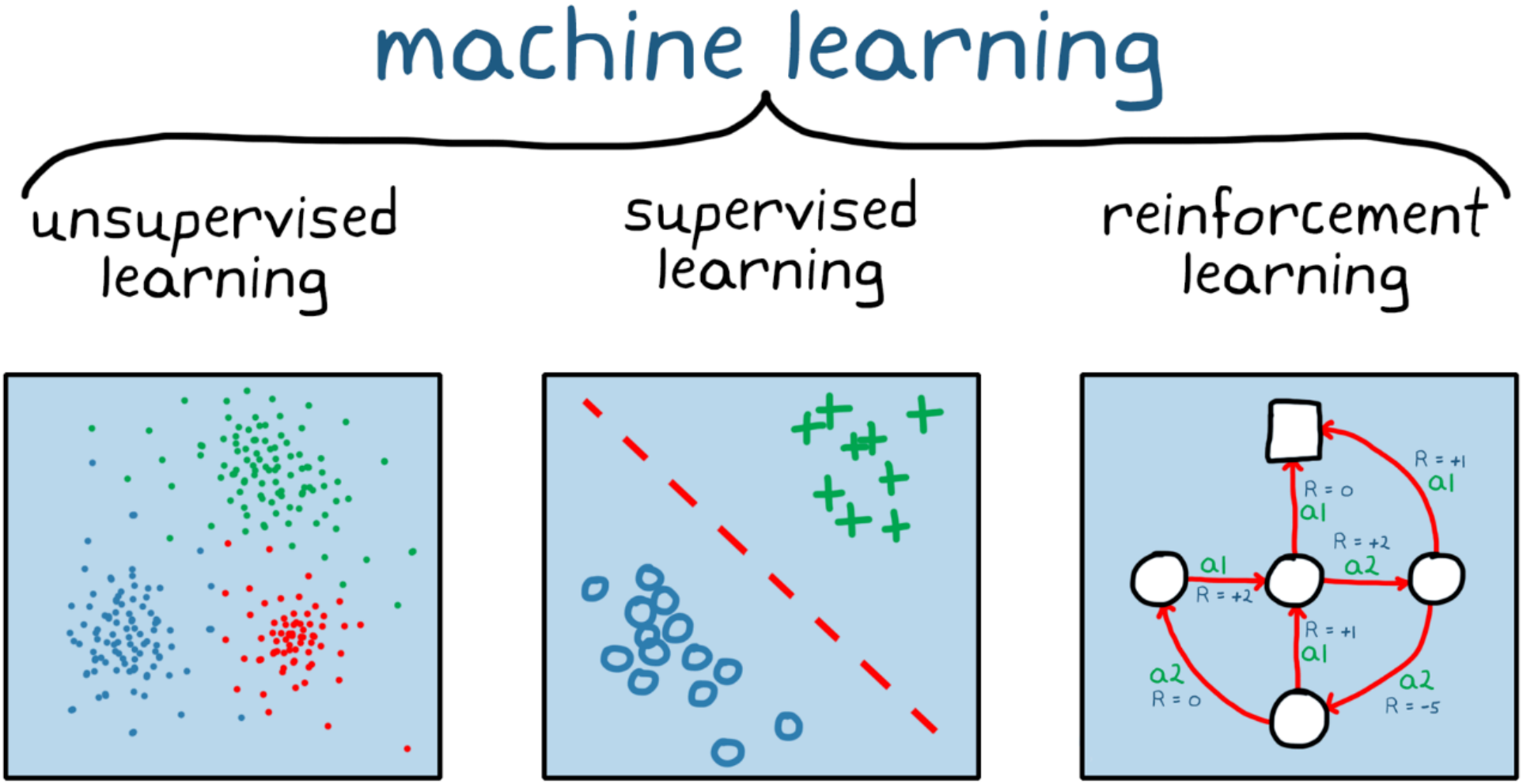
Decision Making and Deep Learning

We will formally define decision making and reinforcement learning later in the course

For now, I want to clarify decision making in the context of machine learning

How does decision making differ from regular deep learning?

Decision Making and Deep Learning



Decision Making and Deep Learning

In deep learning, we usually know the answer

Decision Making and Deep Learning

In deep learning, we usually know the answer

$$f(\boldsymbol{x}, \boldsymbol{\theta}) = \boldsymbol{y}$$

Decision Making and Deep Learning

In deep learning, we usually know the answer

$$f(\boldsymbol{x}, \boldsymbol{\theta}) = \boldsymbol{y}$$

In decision making, we often do not know the answer!

Decision Making and Deep Learning

In deep learning, we usually know the answer

$$f(\boldsymbol{x}, \boldsymbol{\theta}) = \boldsymbol{y}$$

In decision making, we often do not know the answer!

$$f(\boldsymbol{x}, \boldsymbol{\theta}) = ?$$

Decision Making and Deep Learning

In deep learning, we usually know the answer

$$f(\boldsymbol{x}, \boldsymbol{\theta}) = \boldsymbol{y}$$

In decision making, we often do not know the answer!

$$f(\boldsymbol{x}, \boldsymbol{\theta}) = ?$$

What does this mean?

Decision Making and Deep Learning

Example: You train a model f to play chess

Decision Making and Deep Learning

Example: You train a model f to play chess

$$f : X \times \Theta \mapsto Y$$

Decision Making and Deep Learning

Example: You train a model f to play chess

$$f : X \times \Theta \mapsto Y$$

$$X \in \text{Position of pieces on the board}$$

Decision Making and Deep Learning

Example: You train a model f to play chess

$$f : X \times \Theta \mapsto Y$$

$$X \in \text{Position of pieces on the board}$$

$$Y \in \text{Where to move piece}$$

Decision Making and Deep Learning

$X \in$ Position of pieces on the board $\quad$ $Y \in$ Where to move piece

Decision Making and Deep Learning

$X \in$ Position of pieces on the board $\quad Y \in$ Where to move piece

Decision Making and Deep Learning

What is the correct answer?

Decision Making and Deep Learning

What is the correct answer?

We do not know the answer

Decision Making and Deep Learning

How can we learn a model without an answer?

Decision Making and Deep Learning

An answer gives us just one move

Decision Making and Deep Learning

An answer gives us just one move

We need many moves to win

Decision Making and Deep Learning

Decision making can give us the best **sequence** of moves to:

Decision Making and Deep Learning

Decision making can give us the best **sequence** of moves to:

- Win a game of chess

Decision Making and Deep Learning

Decision making can give us the best **sequence** of moves to:

- Win a game of chess
- Drive a customer to the store

Decision Making and Deep Learning

Decision making can give us the best **sequence** of moves to:

- Win a game of chess
- Drive a customer to the store
- Cook a tasty meal

Decision Making and Deep Learning

Decision making can give us the best **sequence** of moves to:

- Win a game of chess
- Drive a customer to the store
- Cook a tasty meal
- Treat a sick patient

Decision Making and Deep Learning

Decision making can give us the best **sequence** of moves to:

- Win a game of chess
- Drive a customer to the store
- Cook a tasty meal
- Treat a sick patient
- Prevent climate change

Decision Making and Deep Learning

Decision making can give us the best **sequence** of moves to:

- Win a game of chess
- Drive a customer to the store
- Cook a tasty meal
- Treat a sick patient
- Prevent climate change
- Reduce human suffering

Decision Making and Deep Learning

Decision making can give us the best **sequence** of moves to:

- Win a game of chess
- Drive a customer to the store
- Cook a tasty meal
- Treat a sick patient
- Prevent climate change
- Reduce human suffering

We do not know the correct moves

Decision Making and Deep Learning

Decision making can give us the best **sequence** of moves to:

- Win a game of chess
- Drive a customer to the store
- Cook a tasty meal
- Treat a sick patient
- Prevent climate change
- Reduce human suffering

We do not know the correct moves

But with decision making, we can find them!

Questions?

Homework

Homework

- Review prerequisites

Homework

- Review prerequisites
  - Especially probability

Homework

- Review prerequisites
  - Especially probability
- Play with Google Colab

Homework

- Review prerequisites
  - Especially probability
- Play with Google Colab
- Download Sutton and Barto textbook

Homework

- Review prerequisites
  - Especially probability
- Play with Google Colab
- Download Sutton and Barto textbook
  - Read Chapter 1.1 (few pages)

Homework

- Review prerequisites
  - Especially probability
- Play with Google Colab
- Download Sutton and Barto textbook
  - Read Chapter 1.1 (few pages)
  - Read Chapter 2 before next lecture